

SPRINGBOK, South Africa

WMO: 685120

Lat: 29.6694S

Lon: 17.8789E

Elev: 1006

StdP: 89.81

Time Zone: 2.00 (E02)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	2.9	3.9	-14.9	1.2	18.8	-12.5	1.4	18.7	15.9	10.0	14.0	10.2	4.4	100	0.486

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	11.7	34.2	16.1	32.9	15.8	31.6	15.5	19.2	28.0	18.3	27.7	17.5	27.4	5.0	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.5	13.3	22.6	15.0	12.0	21.7	13.8	11.1	20.9	59.0	28.1	56.0	27.7	53.2	27.3	23.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.8	10.3	9.1	DB	1.2	37.2	1.0	1.3	0.4	38.1	-0.1	38.9	-0.7	39.6	-1.4	40.6	
			WB	-0.1	20.2	0.9	1.3	-0.8	21.1	-1.3	21.9	-1.7	22.6	-2.4	23.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	17.8	22.6	23.3	22.3	19.7	16.2	12.8	12.4	12.7	15.1	17.5	18.8	21.0
	DBStd	5.92	4.04	3.98	4.16	4.81	4.51	4.03	4.27	4.88	5.22	5.13	4.76	4.06
	HDD10.0	90	0	0	0	1	4	18	26	27	11	3	1	0
	HDD18.3	983	8	5	13	43	96	169	186	185	126	80	52	19
	CDD10.0	2952	390	372	382	292	197	102	101	111	163	236	266	341
	CDD18.3	802	140	143	137	84	31	3	2	11	29	55	67	102
	CDH23.3	7139	1409	1395	1182	588	132	6	6	51	213	503	648	1008
(14) Wind	WSAvg	4.4	4.2	4.0	3.8	4.2	4.3	4.8	5.1	4.5	4.5	4.5	4.5	4.3
	PrecAvg	192	4	7	10	18	21	37	28	27	14	15	7	6
	PrecMax	376	49	65	52	79	96	111	100	94	71	44	52	51
	PrecMin	66	0	0	0	0	0	0	0	1	0	0	0	0
	PrecStd	66	9	12	11	19	24	28	20	25	16	12	13	10
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.8	36.0	34.5	32.6	28.7	24.0	24.1	27.6	31.0	33.5	34.4	34.5
		MCWB	16.7	16.7	16.3	16.0	13.0	10.0	10.4	11.3	12.8	14.6	15.2	15.9
	2%	DB	33.7	33.9	32.7	30.6	26.4	22.2	22.1	24.6	28.2	30.8	31.5	32.5
		MCWB	16.3	16.6	15.9	15.5	12.2	9.9	10.1	10.7	12.1	13.7	14.1	15.2
	5%	DB	31.9	32.2	31.0	28.7	24.7	20.7	20.5	22.5	25.7	28.6	29.5	30.8
		MCWB	16.2	16.4	15.6	14.5	11.9	9.6	9.4	9.8	11.1	13.1	13.6	15.0
	10%	DB	30.0	30.4	29.2	26.8	22.9	18.9	18.8	20.2	23.5	26.2	27.2	28.7
		MCWB	15.5	15.9	14.9	13.8	11.2	8.9	8.8	9.0	10.5	12.4	13.1	14.6
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.1	20.6	20.0	18.7	15.4	13.0	12.1	13.1	15.1	17.2	17.8	18.4
		MCDB	29.5	28.7	27.3	27.5	21.0	17.8	20.8	20.9	24.3	28.6	28.5	29.4
	2%	WB	18.7	19.3	18.6	17.2	14.1	11.5	11.0	11.9	13.7	15.5	16.3	17.3
		MCDB	28.3	27.9	27.1	26.4	21.6	18.1	19.5	21.1	23.6	26.3	26.4	28.2
	5%	WB	17.7	18.4	17.6	16.0	13.2	10.6	10.2	10.9	12.4	14.4	15.1	16.4
		MCDB	27.6	27.5	27.1	25.3	21.3	17.2	18.0	19.8	22.8	25.3	25.5	27.1
	10%	WB	16.7	17.4	16.5	15.1	12.3	9.8	9.4	10.0	11.3	13.3	14.2	15.6
		MCDB	27.0	27.2	26.6	24.1	20.5	16.2	16.8	17.9	21.2	24.0	24.5	26.0
(36) Mean Daily Temperature Range	5% DB	MDBR	12.0	11.7	11.1	9.9	8.6	7.8	8.1	9.2	10.8	11.8	12.2	12.2
		MCDBR	12.8	12.0	11.6	10.7	9.6	9.5	9.4	10.8	12.3	13.1	13.8	13.0
		MCWBR	5.3	5.0	4.8	4.4	3.9	4.3	4.3	4.9	5.4	5.7	5.6	5.5
	5% WB	MCDBR	11.5	10.5	10.1	10.0	8.9	8.4	8.8	10.2	11.7	12.3	12.8	12.1
		MCWBR	5.2	4.7	4.8	4.5	4.1	4.3	4.4	5.0	5.5	5.7	5.9	5.5
(40) Clear-Sky Solar Irradiance	taub		0.300	0.311	0.300	0.284	0.266	0.257	0.262	0.281	0.298	0.307	0.289	0.291
	taud		2.544	2.526	2.547	2.597	2.617	2.607	2.574	2.492	2.413	2.422	2.542	2.555
	Ebn at Noon		1042	1012	992	962	935	923	930	949	977	1005	1046	1054
	Edh at Noon		110	109	100	85	75	71	77	93	112	119	109	109
(44) All-Sky Solar Radiation	RadAvg		8.63	7.85	6.67	5.17	4.00	3.45	3.76	4.63	5.96	7.35	8.36	8.91
	RadStd		0.27	0.23	0.18	0.17	0.15	0.14	0.14	0.17	0.23	0.19	0.25	0.26

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.44	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)	+0.44	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page